

# Math 421: Homework 4

## Solutions

Due: Oct 3, 2025

**Problem 1:** (Ross Chapter 1 4.7) - Let  $S$  and  $T$  be nonempty bounded subsets of  $\mathbb{R}$ .

(a) **Theorem.** If  $S \subseteq T$ , then  $\inf T \leq \inf S \leq \sup S \leq \sup T$ .

*Proof.* Let  $S$  and  $T$  be nonempty bounded subsets of  $\mathbb{R}$ . Then, by completeness,  $\sup S, \sup T, \inf S$ , and  $\inf T$  exist. Suppose  $S \subseteq T$ , then  $\sup T \geq t$  for all  $t \in T$  and thus  $\sup T \geq s$  for all  $s \in S$ , so  $\sup T$  is an upper bound for  $S$  and therefore  $\sup S \leq \sup T$ .

Similarly, we have  $\inf T \leq t$  for all  $t \in T$ , in particular,  $\inf T \leq s$  for all  $s \in S$ . So  $\inf T$  is a lower bound for  $S$ , and therefore  $\inf T \leq \inf S$ .

For every  $s \in S$  we have  $\inf S \leq s$  and  $\sup S \geq s$  so that  $\inf S \leq \sup S$ . Therefore

$$\inf T \leq \inf S \leq \sup S \leq \sup T.$$

□

(b) **Theorem.**  $\sup(S \cup T) = \max\{\sup S, \sup T\}$ .

*Proof.* Let  $S$  and  $T$  be nonempty bounded subsets of  $\mathbb{R}$ . Then  $S \cup T$  is nonempty. Let  $x \in S \cup T$ , if  $x \in S$ , then  $x \leq \sup S \leq \max\{\sup S, \sup T\}$ , if  $x \in T$ , then  $x \leq \sup T \leq \max\{\sup S, \sup T\}$ , so that  $\max\{\sup S, \sup T\}$  is an upper bound for the set  $S \cup T$ . By completeness,  $\sup(S \cup T)$  exists and  $\sup(S \cup T) \leq \max\{\sup S, \sup T\}$ .

Since  $S \subseteq S \cup T$ , we have, by part (a),  $\sup S \leq \sup(S \cup T)$ . Similarly, since  $T \subseteq S \cup T$ , we have  $\sup T \leq \sup(S \cup T)$ . Therefore  $\max\{\sup S, \sup T\} \leq \sup(S \cup T)$ .

Thus, since  $\sup(S \cup T) \leq \max\{\sup S, \sup T\}$  and  $\max\{\sup S, \sup T\} \leq \sup(S \cup T)$ , we have  $\sup(S \cup T) = \max\{\sup S, \sup T\}$ .

□

**Problem 2:** (Ross Chapter 1 4.8) Let  $S$  and  $T$  be nonempty subsets of  $\mathbb{R}$  with the following property:  $s \leq t$  for all  $s \in S$  and  $t \in T$ .

(a) By definition of the sets,  $S$  is bounded above by every element  $t \in T$  and  $T$  is bounded below by every element  $s \in S$ .

(b) **Theorem.**  $\sup S \leq \inf T$ .

*Proof.*  $S$  is nonempty and bounded above by  $t \in T$ , so  $\sup S$  exists. Since every  $t$  is an upper bound for the set  $S$ ,  $\sup S \leq t$  for all  $t \in T$ . Therefore  $\sup S$  is a lower bound for  $T$ , so  $\sup S \leq \inf T$ . □

(c)  $S = \{0, 1\}$ ,  $T = \{1, 2\}$ . Then  $0 \leq 1 \wedge 0 \leq 2$  and  $1 \leq 1 \wedge 1 \leq 2$ .  $S \cap T = \{1\}$ .  $\sup S = 1 = \inf T$ .

(d)  $S = (0, 1)$ ,  $T = (1, 2)$ .  $S \cap T = \emptyset$ .  $\sup S = 1 = \inf T$ .

**Problem 3:** (Ross Chapter 1 4.12)

**Theorem.** Let  $a, b \in \mathbb{R}$ . If  $a < b$ , then there exists an irrational number  $x$  such that  $a < x < b$ .

*Proof.* Let  $a, b \in \mathbb{R}$  such that  $a < b$ . Consider the interval  $(a - \sqrt{2}, b - \sqrt{2})$  by the density of the rational numbers in the real numbers, there exists a rational number  $r \in (a - \sqrt{2}, b - \sqrt{2})$  so that  $a - \sqrt{2} < r < b - \sqrt{2}$  and therefore  $a < r + \sqrt{2} < b$ . We have proven that  $\sqrt{2}$  is irrational and that the sum of a rational number and an irrational number is irrational, so  $x = r + \sqrt{2}$  is irrational and  $a < x < b$ .  $\square$

**Problem 4:** (Ross Chapter 1 4.14 (a)) Let  $A$  and  $B$  be nonempty bounded subsets of  $\mathbb{R}$ , and let  $A + B$  be the set of all sums  $a + b$  where  $a \in A$  and  $b \in B$ .

**Theorem.**  $\sup(A + B) = \sup A + \sup B$

*Proof.* Let  $A$  and  $B$  be nonempty bounded subsets of  $\mathbb{R}$ , and let  $A + B$  be the set of all sums  $a + b$  where  $a \in A$  and  $b \in B$ . Then  $\sup A$  and  $\sup B$  exist by the completeness axiom so that  $a \leq \sup A$  for all  $a \in A$  and  $b \leq \sup B$  for all  $b \in B$ . Then  $a + b \leq \sup A + \sup B$ , thus  $\sup A + \sup B$  is an upper bound for the set  $A + B$ . As  $A + B$  is nonempty as well,  $\sup(A + B)$  exists by completeness.

Let  $M$  be an upper bound of  $A + B$ . Then  $a + b \leq M$  for all  $a + b \in A + B$  which implies  $a \leq M - b$  for all  $a \in A$ . Thus  $M - b$  is an upper bound for  $A$ , so  $\sup A \leq M - b$ . This implies  $b \leq M - \sup A$ , for all  $b \in B$ , so  $M - \sup A$  is an upper bound for  $B$ . Thus  $\sup B \leq M - \sup A$  and therefore  $\sup B + \sup A \leq M$ . Thus  $\sup B + \sup A$  is the least upper bound for  $A + B$ , so,  $\sup(A + B) = \sup A + \sup B$ .  $\square$

**Problem 5:** (Ross Chapter 1 4.15)

**Theorem.** Let  $a, b \in \mathbb{R}$ . If  $a \leq b + \frac{1}{n}$  for all  $n \in \mathbb{N}$ , then  $a \leq b$ .

*Proof.* We prove the contrapositive: Let  $a, b \in \mathbb{R}$ , if  $a > b$ , then there exists  $n \in \mathbb{N}$  such that  $a > b + \frac{1}{n}$ . Since  $a > b$ , we have  $b - a > 0$ . By the Archimedean Property there exists  $n \in \mathbb{N}$  such that  $b - a > \frac{1}{n}$  which implies  $a > b + \frac{1}{n}$ .  $\square$

**Problem 6:** (Ross Chapter 1 4.16)

**Theorem.**  $\sup\{r \in \mathbb{Q} : r < a\} = a$  for each  $a \in \mathbb{R}$ .

*Proof.* Let  $S = \{r \in \mathbb{Q} : r < a\}$ .  $S$  is bounded above by  $a$ , and  $S$  is non-empty, since we can consider the interval  $(a - 1, a)$ , by the density of the rational numbers, there exists  $r \in \mathbb{Q}$  such that  $a - 1 < r < a$  and thus  $r \in S$ . Therefore  $\sup S$  exists.

Let  $\epsilon > 0$ . By the density of the rational numbers in the real numbers, there exists  $r_0 \in \mathbb{Q}$  such that  $r_0 \in (a - \epsilon, a)$ . Thus  $r_0 \in S$ . Therefore for all  $\epsilon > 0$  we have found an  $r_0 \in S$  such that  $a - \epsilon < r_0$ . Thus,  $a = \sup S$ .  $\square$